

M002 Week 7 — Pistons

Topic: Pistons and Mazes · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

A piston opens a path when a lever powers it. Flip the lever, then cross the gap while the path is open.

1 · Predict

Write what happens.

```
move forward
interact ahead
move forward
```

The agent reaches a lever and flips it. A piston opens the path. Can the agent cross now?


```
if gap ahead:
    interact ahead
otherwise:
    move forward
```

The agent is on solid ground, no gap. What does it do?

2 · Trace 🦶

```
keep doing forever:
  if gap ahead:
    interact ahead
    move forward
  otherwise:
    move forward
```

Say in plain English what the agent does at a gap, and what it does on solid ground.

3 · Fill the Code Blank

The agent flips the lever only when it sees a gap ahead.

```
if not agent.detect(BLOCK, FORWARD):  
    agent.____(...)  
    agent.move(FORWARD, 1)  
else:  
    agent.move(FORWARD, 1)
```

Word bank: interact · turn · detect

Write the missing word:

4 · Spot One Bug 🐛

The agent should power the piston **first**, then cross the gap it opens. This code crosses first.

```
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
agent.interact(...)
```

Write the fixed code (flip the lever before the gap):

One sentence: why was it wrong?

5 · Show Your Work 📷 🎥

Open your homework world. The maze has a gap and a piston that opens it. Walk the agent to the lever, flip it, cross while the path is open, and reach the goal.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.